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Problem Statement:

Build a Named Entity Recognition (NER) system for extracting entities from real-world text such as news articles or social media data. And measure its accuracy, precision, recall, and F1 score.

Objective:

- To understand the concept of Named Entity Recognition (NER) and its role in information extraction.
- To load and use a pre-trained BERT-based NER model from Hugging Face Transformers.
- To build a dataset of real-world sentences with annotated gold-standard entity spans.
- To evaluate the NER system using standard metrics: precision, recall, F1 score, and entity accuracy.
- To analyze the performance of the model by comparing predicted spans against gold-standard spans.

S/W Packages and H/W apparatus used:

- Operating System: Windows/Linux/macOS
- Kernel: Python 3.x
- Tools: Jupyter Notebook / Anaconda
- Hardware: CPU with minimum 4GB RAM; GPU recommended for faster inference
- Libraries and packages used: Transformers (Hugging Face), PyTorch (CPU), seqeval, Pandas, Python dataclasses, typing

Theory:

Named Entity Recognition (NER) is a sequence labeling task in NLP that identifies and classifies named entities in text into predefined categories such as Person (PER), Organization (ORG), Location (LOC/GPE), Date (DATE), and Miscellaneous (MISC). NER is a foundational component in information extraction pipelines.

BERT and Transformer-based NER:

BERT (Bidirectional Encoder Representations from Transformers) is a pre-trained deep learning model that captures bidirectional context from text using the Transformer architecture. For NER, BERT is fine-tuned as a token classification model (BertForTokenClassification), where each token is assigned an entity label using the BIO (Beginning-Inside-Outside) tagging scheme. The model used here — `dsllim/bert-base-NER` — is fine-tuned on the CoNLL-2003 NER benchmark dataset and recognizes four entity types: PER, ORG, LOC, and MISC.

Entity Span Representation:

Entities are represented as character-level spans — tuples of (start, end, label) — where start and end are character offsets within the source text. The `aggregation_strategy='simple'` parameter in the Hugging

Face pipeline merges consecutive BIO tokens into single entity spans, making span comparison straightforward.

Evaluation Metrics:

NER systems are evaluated by comparing predicted entity spans against gold-standard annotated spans at the span level (exact match on start, end, and label):

- True Positives (TP): Predicted spans that exactly match a gold span in position and label.
- False Positives (FP): Predicted spans that do not match any gold span.
- False Negatives (FN): Gold spans that were not predicted by the model.
- Precision = $TP / (TP + FP)$: Fraction of predicted entities that are correct.
- Recall = $TP / (TP + FN)$: Fraction of gold entities that were successfully detected.
- F1 Score = $2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$: Harmonic mean balancing precision and recall.
- Entity Accuracy = $TP / (TP + FP + FN)$: Overall proportion of correctly identified entities.

Methodology:

- Install required libraries: transformers, seqeval, and PyTorch (CPU build).
- Load the pre-trained dsliim/bert-base-NER pipeline from Hugging Face with aggregation_strategy='simple' to obtain merged entity spans.
- Define a Span dataclass with fields: start (int), end (int), label (str) to represent annotated entity mentions.
- Create 8 real-world sample sentences covering news, sports, and social media topics with manually annotated gold entity spans for entity types: ORG, GPE, PERSON, DATE, and EVENT.
- Convert mention strings to character-level Span objects using sequential text search to locate each mention's exact start and end offsets.
- Run the NER pipeline on a sample sentence and display predicted entities with their entity group, character offsets, and confidence scores.
- Implement an evaluate() function that iterates over all samples, computes TP, FP, and FN by comparing predicted and gold span sets, and calculates precision, recall, F1 score, and entity accuracy.
- Run the evaluation function on the full dataset and display the resulting metrics dictionary.

Use Cases:

- Use Case 1 - News Article Analysis: NER systems automatically extract persons, organizations, locations, and dates from news feeds, enabling automated knowledge graph construction and event tracking in media monitoring platforms.
- Use Case 2 - Social Media Intelligence: Identifying entities such as public figures, brands, and locations in tweets and posts helps organizations monitor brand mentions, track trending topics, and detect misinformation in real time.
- Use Case 3 - Legal and Financial Document Processing: NER extracts party names, dates, monetary amounts, and jurisdiction names from contracts and financial reports, automating information extraction for compliance and due diligence workflows.

Advantages:

- Pre-trained BERT-based models provide high accuracy on real-world text without requiring training data.
- The Hugging Face pipeline abstracts tokenization, inference, and span aggregation into a single API call.
- Character-level span evaluation provides strict, unambiguous assessment of model performance.
- The modular `evaluate()` function is reusable across different datasets and NER models.
- BERT's bidirectional context allows it to resolve entity ambiguity better than traditional rule-based or CRF-based NER systems.

Limitations:

- The `dslim/bert-base-NER` model only recognizes four entity types (PER, ORG, LOC, MISC); custom types like DATE and EVENT in the gold annotations are not supported, reducing exact-match scores.
- Strict span-level evaluation penalizes partial matches; a prediction that identifies the correct entity but with slightly different boundaries counts as both a FP and FN.
- BERT inference on CPU is slow; processing large corpora requires a GPU for practical deployment.
- The model may underperform on informal text (social media slang, abbreviations) as it was fine-tuned on formal news data (CoNLL-2003).
- Small evaluation datasets (8 sentences) may not provide statistically representative performance estimates.

Applications:

- Information retrieval: NER-enriched search engines return results filtered by entity type (e.g., all articles mentioning a specific person or organization).
- Question answering systems: Entity extraction identifies candidate answers (names, dates, locations) for factoid questions.
- Automated resume parsing: NER extracts skills, organizations, job titles, and education details from candidate CVs.
- Healthcare: Clinical NER extracts patient names, drug names, diagnoses, and dosage information from medical records.
- Financial analytics: Extracting company names, financial figures, and dates from earnings reports and news for algorithmic trading signals.

Conclusion:

This assignment implemented a Named Entity Recognition system using the pre-trained `dslim/bert-base-NER` model from Hugging Face Transformers. Eight real-world sentences were annotated with gold entity spans and evaluated against the model's predictions using precision, recall, F1 score, and entity accuracy. The results highlighted the gap between the model's CoNLL-2003 entity types (PER, ORG, LOC, MISC) and the gold annotations which included custom types like DATE and EVENT, demonstrating how schema mismatch affects strict span-level evaluation. The exercise provided practical exposure to transformer-based NER pipelines and the standard evaluation methodology used in NLP benchmarks.